

2023



Pioneering the Connected Vehicle Cockpit Using AWS IoT Services with Visteon

Learn how Visteon is helping automakers meet driver expectations for cockpit functionality with solutions built using AWS and AWS IoT services.

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Overview

As drivers increasingly expect their cars to connect seamlessly with their digital lives, [Visteon](#) is redefining how it builds cockpit electronics for automakers. “We started our journey by looking at what has made smartphones successful,” says Sivakumar Yeddanapudi, Visteon’s vice president of product management, digital cockpit electronics. “Visteon applies the smartphone experience to how users interact with their cars.”

To develop its connected vehicle innovations, Visteon builds its suite of solutions—called AllGo—on top of Amazon Web Services (AWS). AllGo uses AWS services for storage, compute, Internet of Things (IoT) functionality, and vehicle connectivity. Using AWS, Visteon scales to accommodate millions of vehicles, iterates quickly, and continues to enhance the driver experience.



Opportunity | Connecting Millions of Vehicles to the Cloud Using AWS IoT Core

Based in Michigan, Visteon provides in-vehicle technology to automakers around the globe, helping them bring digital products to drivers quickly without it requiring millions of dollars in development costs.

Since introducing the industry’s first on-board domain controller in 2018, Visteon has experienced exponential growth—with the third quarter of 2023 marking 18 consecutive quarters in which it has outperformed the market. The domain controller, Visteon SmartCore, consolidates a vehicle’s electronic control units (ECUs). The ECUs use software, graphics processors, and CPUs powered by [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. To securely connect vehicles to the cloud, Visteon uses [AWS IoT Core](#), which lets companies connect billions of IoT devices and route trillions of services without managing infrastructure.

Using AWS IoT Core and other AWS services, Visteon can help automakers provide value for customers and address their own challenges. For example, to update or repair on-board software, automakers typically spend millions of dollars on widespread service recalls. Instead, Visteon wants to provide wirelessly delivered—over-the-air (OTA)—updates. “This could help reduce warranty cost, manage some regulatory issues, and handle new cybersecurity threats,” says Sivakumar.

Solution | Iterating Quickly to Build Innovative Connected Vehicle Solutions

To develop their connected vehicle solutions, small internal teams within Visteon iterate in independent AWS environments using built-in security features that incorporate industry standards. “We work like a startup, creating quick prototypes by expanding on available AWS services based on user requirements,” says Pradeep Balakrishnan, Visteon’s senior engineering manager. “Our productivity increases, and we deliver innovative products faster.” Alongside AWS solutions architects and technical account managers, Visteon operates in line with the [AWS Well-Architected Framework](#), which describes key concepts, design principles, and architectural best practices for designing and running workloads in the cloud.

In 2022, Visteon unveiled plans for its AllGo App Store, which features preloaded popular apps for navigation, parking, infotainment, and other functions. To store files from app developers, Visteon uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering industry-leading scalability, data availability, security, and performance. In fact, Visteon uses Amazon S3 for all of its object storage, including differential images and binary software images that are important for OTA updates.

For the OTA update feature, Visteon assigns a unique identity to each vehicle using AWS IoT Core’s [device registry](#). Using [AWS IoT Device Management](#)—which helps register, organize, monitor, and remotely manage IoT devices at scale—Visteon groups devices to receive automatic updates according to specific criteria. The automaker sets the speed of deployment as well as campaign start and end dates using [Jobs](#), a feature of AWS IoT Device Management, which defines a set of remote operations that can be sent to and run on one or more devices connected to AWS IoT Core. “We can point to every vehicle that’s available with the automaker and monitor the state of each campaign and every detail seamlessly in the cloud,” Balakrishnan says.



Additionally, Visteon built a subscription management system that lets users subscribe to weather or traffic updates or other third-party services from inside the vehicle. In effect, the system acts as an authorization provider, managing details such as payments and expiration dates. A North American automaker has already made the system a feature in its cars, thus giving Visteon an additional revenue source. Visteon is also developing a state-of-the-art vehicle security mode called Scout, which streams video from a vehicle's camera to the owner's mobile device for remote video monitoring. The solution uses [Amazon Kinesis Video Streams](#), which makes it simple to securely stream video from connected devices to AWS for analytics, machine learning, playback, and other processing.

Outcome | Continuing to Innovate in the Automotive Industry

Using AWS services, Visteon continues to position itself as an innovator in the automotive industry, with plans to widely release its OTA update services and app store by 2025. It also is working on new use cases focused on vehicle analytics and remote commands.

"AWS is constantly thinking about automotive problems and how to use the power of the cloud to accelerate automotive transformation," says Sivakumar. "Using AWS IoT Core and other services helps us innovate with a company that has strong automotive intellect."

About Visteon

Based in Michigan, Visteon is a global technology company serving the mobility industry, dedicated to creating a more enjoyable, connected, and convenient driving experience.

AWS Services Used

AWS IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)

AWS IoT Device Management

AWS IoT Device Management helps you register, organize, monitor, and remotely manage scale.



[Learn more »](#)

Amazon Kinesis Video Streams

Amazon Kinesis Video Streams makes it easy to securely stream video from connected devices to AWS for analytics, machine learning (ML), playback, and other processing.

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Amazon EC2

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EUROPE, MIDDLE EAST, & AFRICA



Continental on AWS

From one of the most recognized tire brands in the world to a connected mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to the cloud and provides a virtual workbench to speed development from



2024

2024

ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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